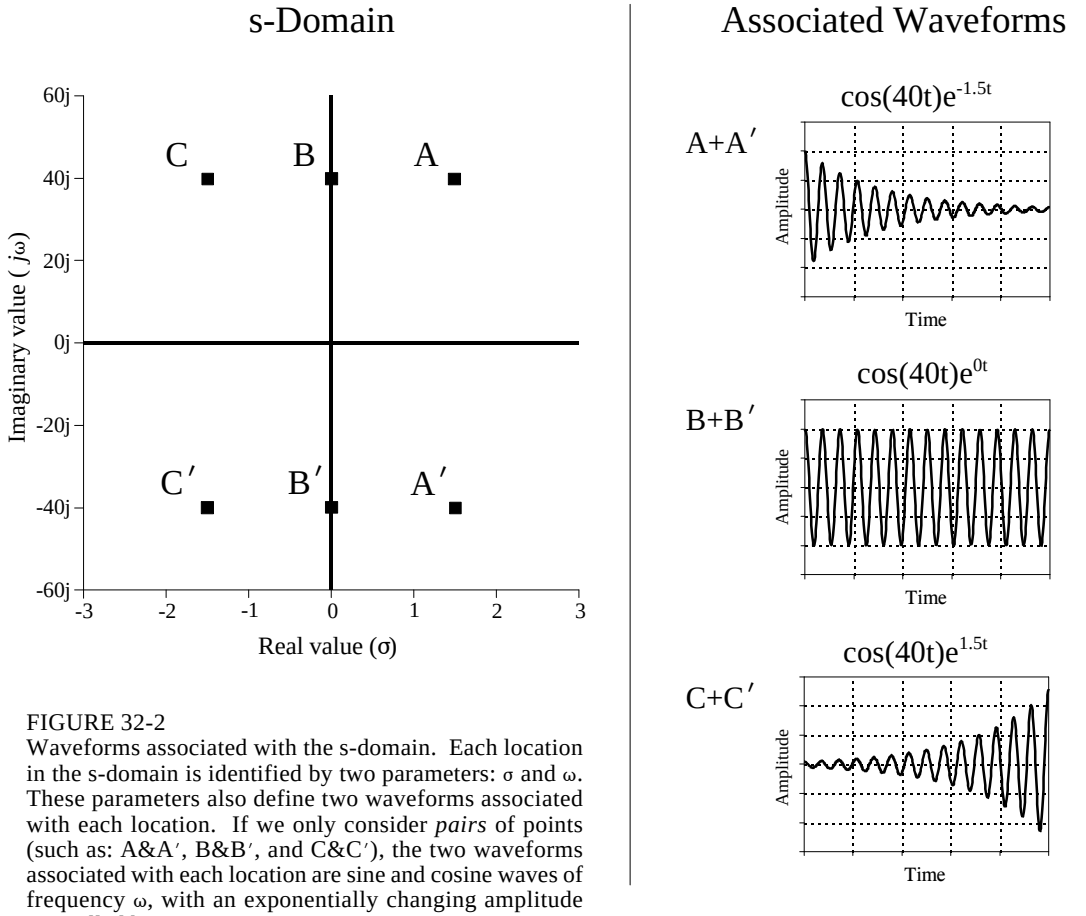



FIGURE 32-2
Waveforms associated with the s-domain. Each location in the s-domain is identified by two parameters: σ and ω. These parameters also define two waveforms associated with each location. If we only consider *pairs* of points (such as: A&A′, B&B′, and C&C′), the two waveforms associated with each location are sine and cosine waves of frequency ω, with an exponentially changing amplitude controlled by σ.

Figure 32-2 shows three *pairs* of points in the s-plane: A&A′, B&B′, and C&C′. Just as in the complex frequency spectrum, the points at A, B, & C (the positive frequencies) are the complex conjugates of the points at A′, B′, & C′ (the negative frequencies). The top half of the s-plane is a mirror image of the lower half, and both halves are needed to correspond with a real time domain signal. In other words, treating these points in pairs bypasses the complex math, allowing us to operate in the time domain with only real numbers.

Since each of these pairs has specific values for σ and ±ω, there are two waveforms associated with each pair: $\cos(\omega t)\,e^{-\sigma t}$ and $\sin(\omega t)\,e^{-\sigma t}$. For instance, points A&A′ are at a location of $\sigma = 1.5$ and $\omega = \pm 40$, and therefore correspond to the waveforms: $\cos(40t)\,e^{-1.5t}$ and $\sin(40t)\,e^{-1.5t}$. As shown in Fig. 32-2, these are sinusoids that exponentially *decreases* in amplitude as time progresses. In this same way, the sine and cosine waves associated with B&B′ have a *constant* amplitude, resulting from the value of σ being zero. Likewise, the sine and cosine waves that are associated with locations C&C′ exponentially *increases* in amplitude, since σ is negative.